Oleh Savchuk

Curriculum Vitae

Research interests

Heavy-Ion Collisions; QCD Equation of State; Gravity; Machine Learning; Data Science.

Academic appointments

Nov 2025 -··· **Adjunct Research Associate**, *Facility for Rare Isotope Beams, MI, USA*.
Dense Nuclear Matter Equation of State

2024-2025 **Research Associate**, *Facility for Rare Isotope Beams, MI, USA*.
Dense Nuclear Matter Equation of State

2022-2024 **Visiting Researcher**, *Facility for Rare Isotope Beams, MI, USA*.
Fluctuations and correlations in heavy-ion collisions

Summer 2023 **Visiting Researcher**, *CERN, Switzerland*.
Testing experimental sensitivity at new experiments to dark matter candidates

2022 **Visiting Researcher**, *GSI, Darmstadt*.
Equation of state at SIS18/SIS100 with flow, dileptons, fluctuations

2020-2021 **Research Scholar**, *Frankfurt Institute for Advanced Studies, Germany*.
Application of machine learning and artificial intelligence to statistical physics

Awards

2024 **International Research Travel Award Program – Ukraine**.
For a 1-month stay at Lawrence Berkeley National Laboratory to collaborate with Dr. Volker Koch

Education

2021–2024 **PhD student**, *Bogolyubov Institute for Theoretical Physics, Kyiv, Ukraine*,
Thesis: Equation of State of Strongly Interacting Matter and Relativistic Heavy-Ion Collisions,
Advisor: Prof. Dr. Mark Gorenstein.

2018–2020 **Master student**, *Theoretical Physics,Taras Shevchenko National University of Kyiv, Kyiv, Ukraine*,
Advisor: Prof. Dr. Mark Gorenstein.

2014–2018 **Bachelor student**, *Theoretical Physics, Taras Shevchenko National University of Kyiv, Kyiv, Ukraine*,
Thesis: Finite Volume and Lifetime Effects on Secondary-Particle Spectra in Relativistic Heavy-Ion Collisions,
Advisor: Prof. Dr. Dmitry Anchishkin.

Computer Skills

Programming C++, Python, Pybind11, Java, C, Fortran, OpenMP, MPI, CUDA

Software UrQMD, SMASH, Mathematica, ROOT, Keras, TensorFlow, LaTeX, Maple, Matlab, COMSOL Multiphysics, HyperChem, Visual Molecular Dynamics

Other Data Analysis and Visualization, Numerical methods, Monte Carlo

Academic output

INSPIRE metrics Total Citations: 529 , h-index 11

Publications 23 on QCD properties, dynamics and equation of state. 2 papers on general relativity. 1 paper on Machine Learning and Normalizing flows.

Talks 19 Presentations and 1 poster session on international conferences.

Teaching Introductory courses in calculus, various seminars and lectures on phenomenology of heavy-ion collisions, mathematical physics, numerical modelling for students of Taras Shevchenko National University, Kyiv Academic University and Bogolyubov Institute for Theoretical Physics.

Outreach

2025 Volunteered to help orginize FRIB stand at MSU science fair.

2025 Volunteered to help present FRIB science to school students.

2024 Volunteered to help organize "Isotopes in motion" play at FRIB

2024 Volunteered to help present FRIB science at WKAR and PBS Kids day.

2024 Participated in GeMS luncheon at DNP

2023 Occasional seminars at BITP for general audiences

2022 Helped Ukrainian graduate students find safe places abroad

2018-2020 Gave supplementary lessons in Calculus and Mechanics to first year university students

2016-2020 Helped to run table-top games club at Taras Shevchenko National University

Publications

[1] O. Savchuk, “Net-proton fluctuations influenced by baryon stopping and quark deconfinement”, Phys. Rev. C **111**, 024913 (2025), arXiv:`2407.17670 [hep-ph]`.

[2] O. Savchuk, “Sensitivity of transverse momentum correlations to early-stage and thermal fluctuations”, J. Phys. G **52**, 035106 (2025), arXiv:`2402.12504 [hep-ph]`.

[3] O. Savchuk, and S. Pratt, “Correlations of conserved quantities at finite baryon density”, Phys. Rev. C **109**, 024910 (2024), arXiv:2311.02046 [nucl-th].

[4] O. S. Stashko, O. V. Savchuk, and V. I. Zhdanov, “Quasinormal modes of naked singularities in presence of nonlinear scalar fields”, Phys. Rev. D **109**, 024012 (2024), arXiv:2307.04295 [gr-qc].

[5] V. A. Kuznietsov, O. Savchuk, R. V. Poberezhnyuk, V. Vovchenko, M. I. Gorenstein, and H. Stoecker, “Molecular dynamics analysis of particle number fluctuations in the mixed phase of a first-order phase transition”, Phys. Rev. C **107**, 055206 (2023), arXiv:2303.09193 [hep-ph].

[6] O. S. Stashko, O. V. Savchuk, L. M. Satarov, I. N. Mishustin, M. I. Gorenstein, and V. I. Zhdanov, “Pion stars embedded in neutrino clouds”, Phys. Rev. D **107**, 114025 (2023), arXiv:2303.06190 [hep-ph].

[7] T. Reichert, O. Savchuk, A. Kittiratpattana, P. Li, J. Steinheimer, M. Gorenstein, and M. Bleicher, “Decoding the flow evolution in Au+Au reactions at 1.23A GeV using hadron flow correlations and dileptons”, Phys. Lett. B **841**, 137947 (2023), arXiv:2302.13919 [nucl-th].

[8] A. Sorensen, et al., “Dense nuclear matter equation of state from heavy-ion collisions”, Prog. Part. Nucl. Phys. **134**, 104080 (2024), arXiv:2301.13253 [nucl-th].

[9] O. Savchuk, R. V. Poberezhnyuk, A. Motornenko, J. Steinheimer, M. I. Gorenstein, and V. Vovchenko, “Phase transition amplification of proton number fluctuations in nuclear collisions from a transport model approach”, Phys. Rev. C **107**, 024913 (2023), arXiv:2211.13200 [hep-ph].

[10] S. Chen, O. Savchuk, S. Zheng, B. Chen, H. Stoecker, L. Wang, and K. Zhou, “Fourier-flow model generating Feynman paths”, Phys. Rev. D **107**, 056001 (2023), arXiv:2211.03470 [hep-lat].

[11] O. Savchuk, A. Motornenko, J. Steinheimer, V. Vovchenko, M. Bleicher, M. Gorenstein, and T. Galatyuk, “Enhanced dilepton emission from a phase transition in dense matter”, J. Phys. G **50**, 125104 (2023), arXiv:2209.05267 [nucl-th].

[12] O. Savchuk, R. V. Poberezhnyuk, and M. I. Gorenstein, “Possible origin of HADES data on proton number fluctuations in Au+Au collisions”, Phys. Lett. B **835**, 137540 (2022), arXiv:2209.01883 [hep-ph].

[13] M. Gazdzicki, M. Gorenstein, I. Pidhurskyi, O. Savchuk, and L. Tinti, “Equilibration and Locality”, Acta Phys. Polon. B **53**, 8 (2022), arXiv:2206.01151 [nucl-th].

[14] V. A. Kuznietsov, O. Savchuk, O. S. Stashko, and M. I. Gorenstein, “Critical point influenced by Bose-Einstein condensation”, Phys. Rev. C **106**, 034319 (2022), arXiv:2206.00424 [hep-ph].

[15] V. A. Kuznietsov, O. Savchuk, M. I. Gorenstein, V. Koch, and V. Vovchenko, “Critical point particle number fluctuations from molecular dynamics”, Phys. Rev. C **105**, 044903 (2022), arXiv:2201.08486 [hep-ph].

[16] V. A. Kuznietsov, O. S. Stashko, O. V. Savchuk, and M. I. Gorenstein, “Critical point and Bose-Einstein condensation in pion matter”, Phys. Rev. C **104**, 055202 (2021), arXiv:2108.08140 [hep-ph].

[17] O. Savchuk, V. Vovchenko, V. Koch, J. Steinheimer, and H. Stoecker, "Constraining baryon annihilation in the hadronic phase of heavy-ion collisions via event-by-event fluctuations", Phys. Lett. B **827**, 136983 (2022), arXiv:2106.08239 [hep-ph].

[18] O. S. Stashko, O. V. Savchuk, R. V. Poberezhnyuk, V. Vovchenko, and M. I. Gorenstein, "Phase diagram of interacting pion matter and isospin charge fluctuations", Phys. Rev. C **103**, 065201 (2021), arXiv:2102.02567 [hep-ph].

[19] R. V. Poberezhnyuk, O. Savchuk, M. I. Gorenstein, V. Vovchenko, and H. Stoecker, "Higher order conserved charge fluctuations inside the mixed phase", Phys. Rev. C **103**, 024912 (2021), arXiv:2011.06420 [hep-ph].

[20] M. Gazdzicki, M. I. Gorenstein, O. Savchuk, and L. Tinti, "Notes on statistical ensembles in the Cell Model", Int. J. Mod. Phys. E **29**, 2050060 (2020).

[21] O. S. Stashko, D. V. Anchishkin, O. V. Savchuk, and M. I. Gorenstein, "Thermodynamic properties of interacting bosons with zero chemical potential", J. Phys. G **48**, 055106 (2021), arXiv:2007.06321 [hep-ph].

[22] R. V. Poberezhnyuk, O. Savchuk, M. I. Gorenstein, V. Vovchenko, K. Taradiy, V. V. Begun, L. Satarov, J. Steinheimer, and H. Stoecker, "Critical point fluctuations: Finite size and global charge conservation effects", Phys. Rev. C **102**, 024908 (2020), arXiv:2004.14358 [hep-ph].

[23] O. Savchuk, Y. Bondar, O. Stashko, R. V. Poberezhnyuk, V. Vovchenko, M. I. Gorenstein, and H. Stoecker, "Bose-Einstein condensation phenomenology in systems with repulsive interactions", Phys. Rev. C **102**, 035202 (2020), arXiv:2004.09004 [hep-ph].

[24] V. Vovchenko, O. Savchuk, R. V. Poberezhnyuk, M. I. Gorenstein, and V. Koch, "Connecting fluctuation measurements in heavy-ion collisions with the grand-canonical susceptibilities", Phys. Lett. B **811**, 135868 (2020), arXiv:2003.13905 [hep-ph].

[25] O. Savchuk, R. V. Poberezhnyuk, V. Vovchenko, and M. I. Gorenstein, "Binomial acceptance corrections for particle number distributions in high-energy reactions", Phys. Rev. C **101**, 024917 (2020), arXiv:1911.03426 [hep-ph].

[26] O. Savchuk, V. Vovchenko, R. V. Poberezhnyuk, M. I. Gorenstein, and H. Stoecker, "Traces of the nuclear liquid-gas phase transition in the analytic properties of hot QCD", Phys. Rev. C **101**, 035205 (2020), arXiv:1909.04461 [hep-ph].

Talks

1. Efficient Emulation of Smooth Functions for Bayesian Inference, APS DNP, Chicago 2025
2. Early-stage effects in fluctuations and correlations of conserved charges, APS DNP, Chicago 2025
3. Femtoscopy of rotating sources, APS DNP, Chicago 2025
4. Femtoscopy of rotating sources, XV Conference of Young Scientists, BITP, Kyiv 2025 Problems of Theoretical Physics
5. Deconfinement effects in net-proton fluctuations, 15th Conference on the Intersections of Particle and Nuclear Physics, 2025
6. Initial state and recombination effects on net-proton fluctuation, Journal club at Vanderbilt University, 2025
7. Net-proton fluctuations influenced by baryon stopping and quark deconfinement, WPCF 2024
8. Minimizing Emulator Uncertainty, October 28, 2024 to November 1, 2024 IRL-NPA FRIB

9. Minimizing Emulator Uncertainty, 2024 Fall Meeting of the APS Division of Nuclear Physics
10. Baryon Stopping Signal in Net-Proton Fluctuations, 2024 Fall Meeting of the APS Division of Nuclear Physics
11. Fluctuations and local charge conservation,CERN - Ukraine 2024: “Past - Present - Future” Conference, Kyiv Ukraine, May 28-29, 2024
12. Enhanced dilepton emission from a phase transition in dense matter, Dense Nuclear Matter Equation of State from Heavy-Ion Collisions, Seattle, WA, Dec 5-9, 2022
13. Correlations of conserved charges at finite density Quark Matter 2023, Houston, Texas, Sep 3-9, 2023
14. Dilepton production from coarse grained UrQMD with a CMF equation of state, Transport Meeting, Frankfurt, Germany, April 14th, 2022
15. Enhanced dilepton emission from a phase transition in dense matter, BES-Tea Seminar Series, September 7, 2022, Online
16. Constraining baryon annihilation in the hadronic phase of heavy-ion collisions via event-by-event fluctuations, Heavy Ion Collisions in the QCD phase diagram, Subatech, Nantes, France, June 26th - July 9th, 2022
17. Constraining baryon annihilation in the hadronic phase of heavy-ion collisions via event-by-event fluctuations, 12h Conference of young scientists “Problems of theoretical physics”, Kyiv, Ukraine, 21-22 December 2021
18. Bose-Einstein condensation phenomenology in systems with repulsive interactions, 11th Conference of young scientists “Problems of theoretical physics”, Kyiv, Ukraine, 21-23 December 2020
19. Finite Volume and Lifetime Effects on Secondary-Particle Spectra in Relativistic Collisions, 18th Zimanyi School winter workshop on Heavy-Ion Collisions, Budapest, Hungary, December 3-7th, 2018
20. Traces of nuclear liquid-gas transition in analytic structure of hot QCD, 19th Zimanyi School winter workshop on Heavy-Ion Collisions, Budapest, Hungary, December 3-7th, 2019

Languages

Ukrainian	native
English	fluent